

Electrical Circuits I

ENGR 2405

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Chapter 03 – Methods of Analysis

Austin Community College

Overview

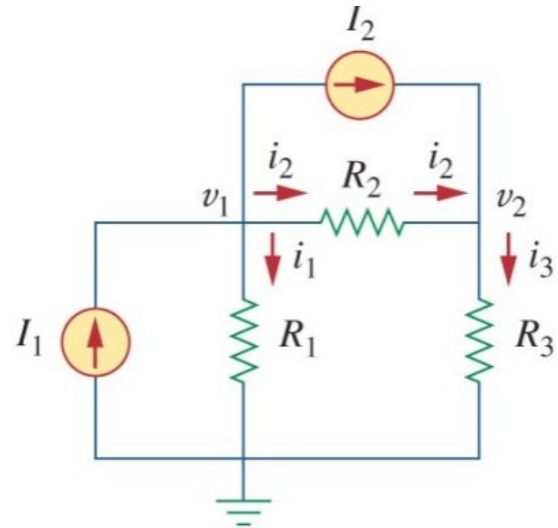
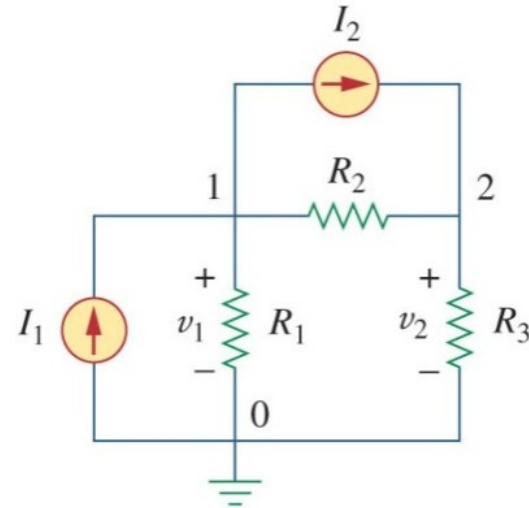
- Nodal Analysis based on KCL
- Mesh Analysis based on KVL
- Simultaneous Equations
- LTSpice

Nodal Analysis (NA)

- Three Step Process:
 1. Select a node as the reference node (v_o) and assign voltages $v_1, v_2, \dots, v_{n-1}$ to the remaining nodes
 2. Apply KCL to each of the $n-1$ non-reference nodes and use Ohm's law to express branch currents in terms of node voltages
 3. Solve the resulting $n-1$ simultaneous equations to obtain the unknown node voltages
- The reference node (v_o) is commonly referred to as "ground" as its voltage, by default, is zero

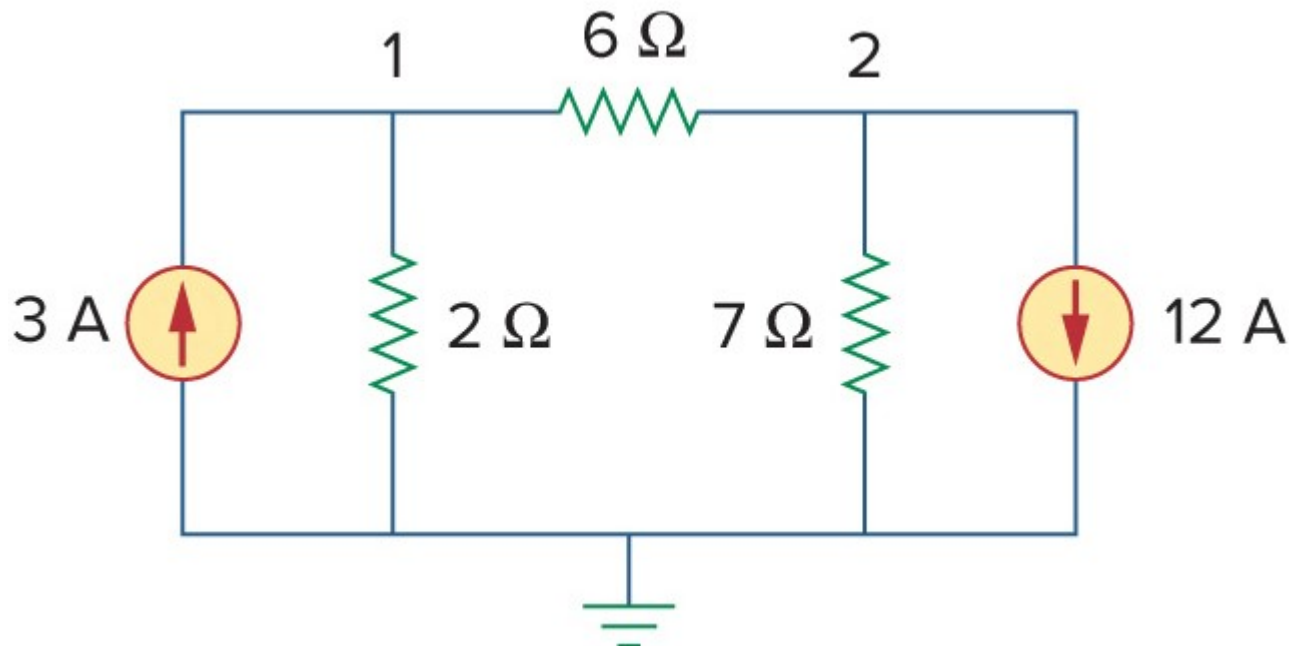
Applying NA - Example

- Use Ohm's law to express the currents in terms of node voltages, keeping in mind that current flows from high potential to low
- Substitute back into the node current equations
- Resulting in 2 equations with 2 unknowns

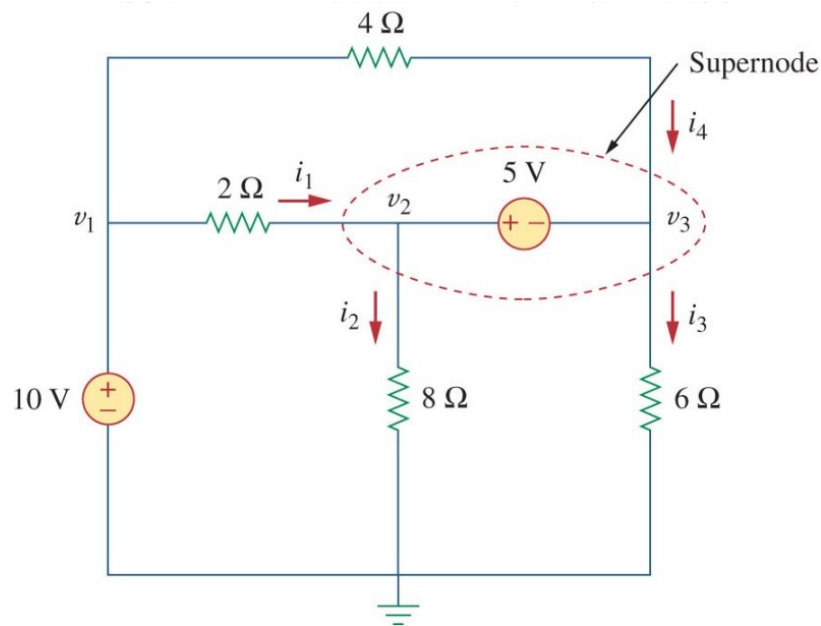


Practice Problems

3.1 Obtain the node voltages



Including Voltage Sources



Depending on where the V source is located, the approach varies:

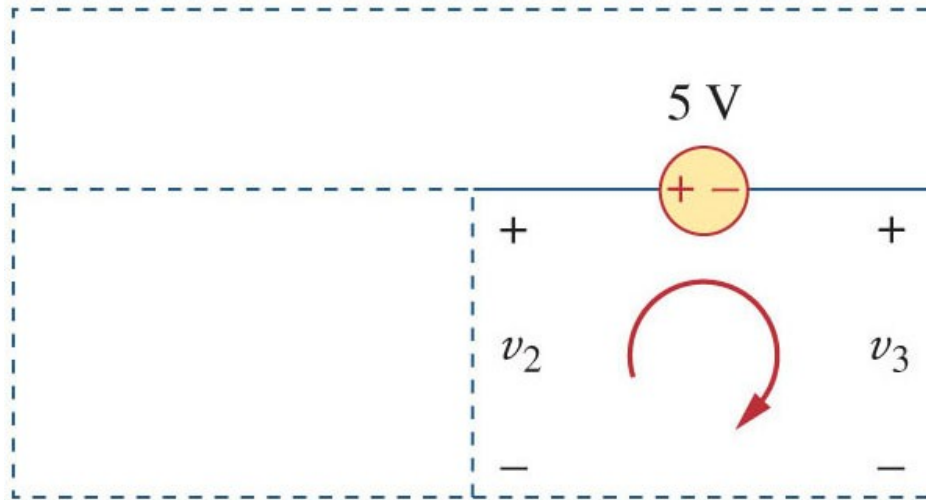
- Between v_o and v_i : set the $v_i = V_s$ (i.e. set $v_1 = 10\text{V}$)
- Between v_j and v_k : the two nodes form a "supernode"

Supernode

- A supernode is formed by enclosing the voltage source connected between two non-reference nodes (and any elements in parallel with it)
- As NA requires using KCL and we don't know what the current through a voltage source in advance, we need to take this approach
- By lumping the nodes together, the current balance can still be described as before
- In the previous circuit, nodes 2 and 3 form a supernode

Supernode

- Looking at just the supernode:



- We can use KVL to create a "constraint" equation

Equations to Solve

- 3 Equations with 3 Unknowns

$$v_1 = 10$$

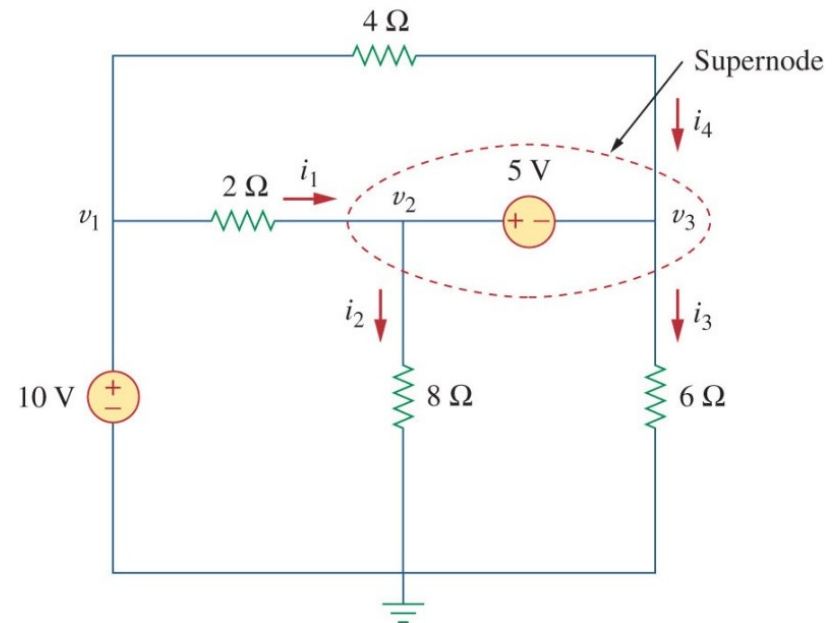
$$v_2 - v_3 = 5 \text{ (from supernode)}$$

$$\frac{v_1 - v_2}{2} + \frac{v_1 - v_3}{4} = \frac{v_2 - 0}{8} + \frac{v_3 - 0}{6}$$

- Really 2 Equations with 2 Unknowns

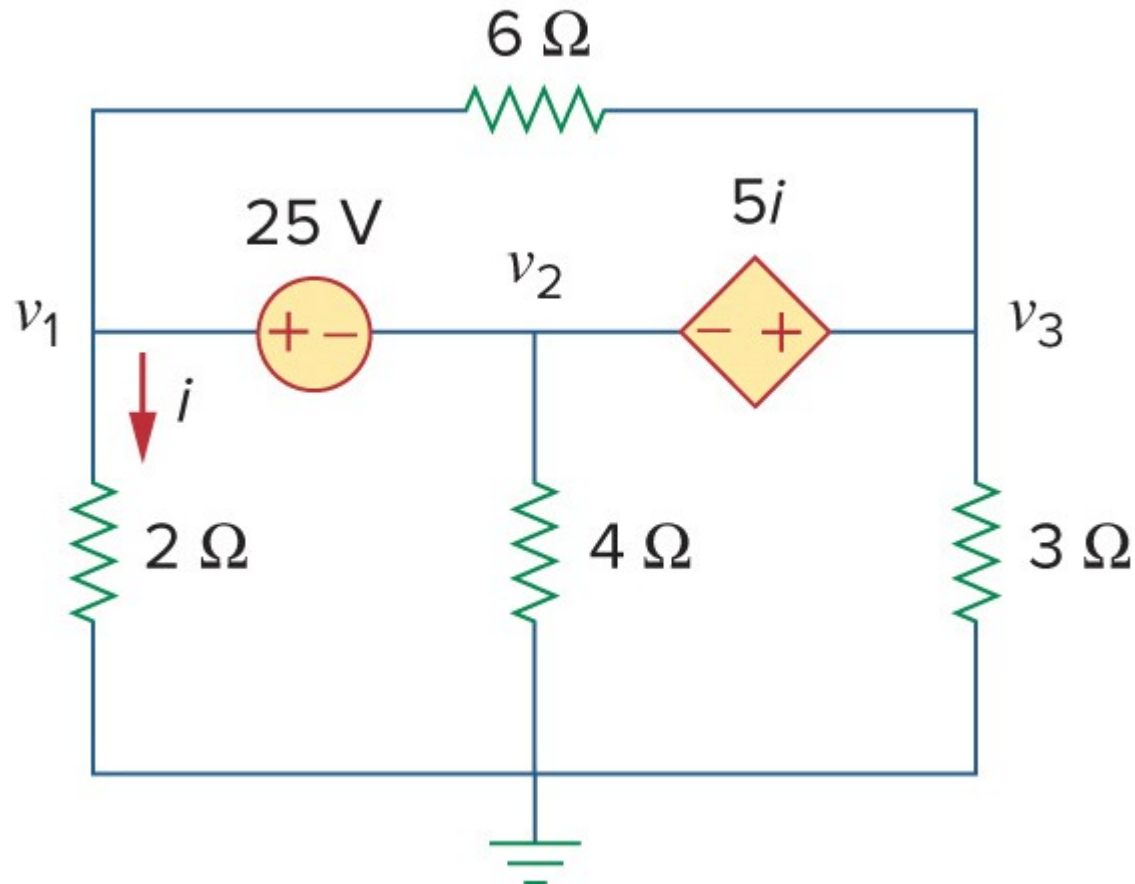
$$v_2 - v_3 = 5$$

$$15v_2 + 10v_3 = 180$$



Practice Problems

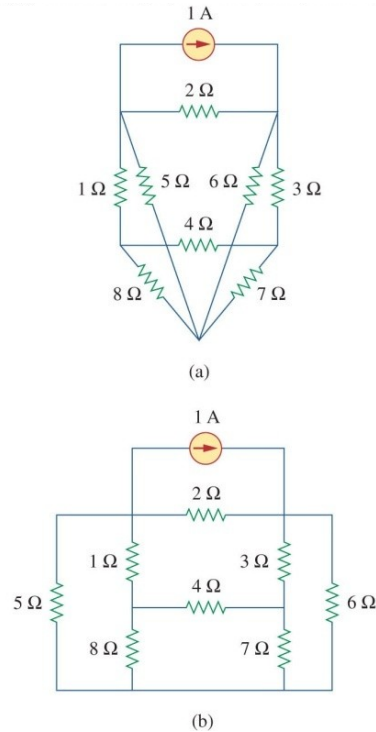
3.4 Find v_1 , v_2 , and v_3



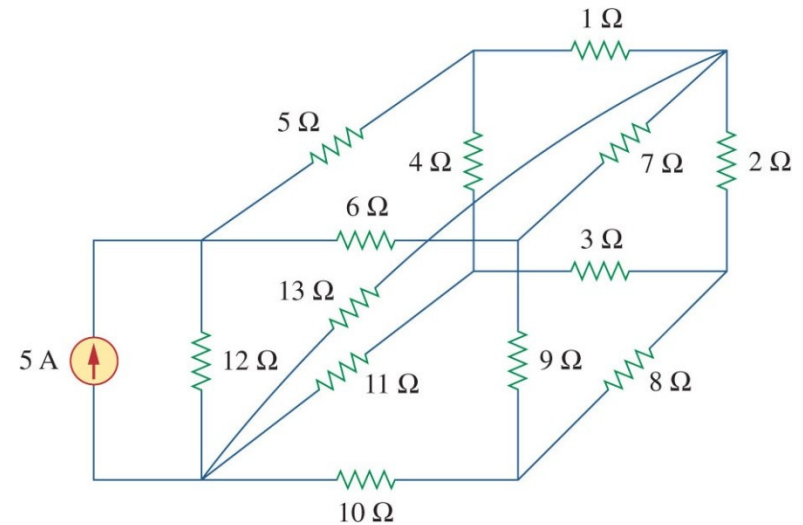
Mesh Analysis (MA)

- Another general procedure for analyzing circuits is to use the mesh currents as the circuit variables (mesh is a loop without any other loop within it)
- MA uses KVL to find unknown currents
- MA is limited in one aspect: can only apply to planar circuits
- A planar circuit is one that can be drawn such that there are no crossing branches

Planar versus Non-Planar

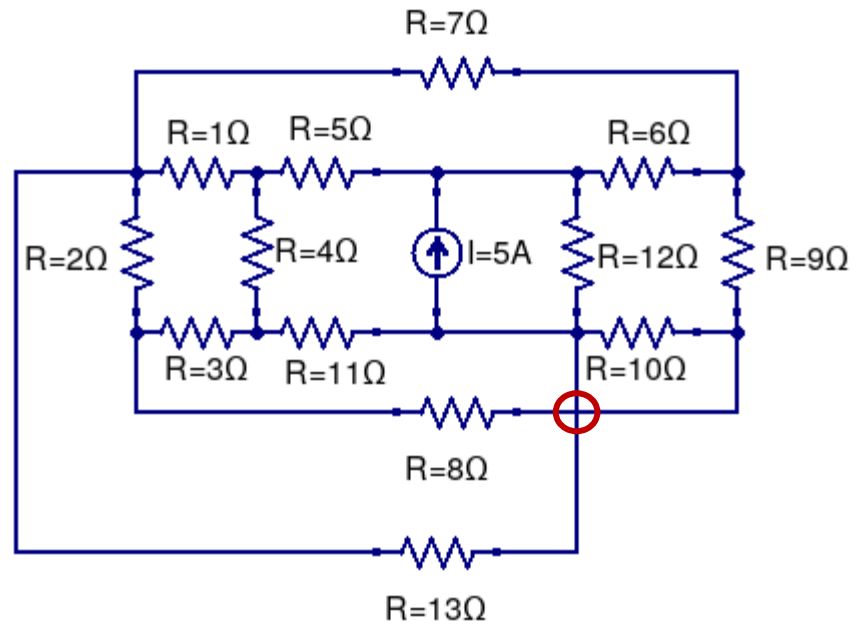
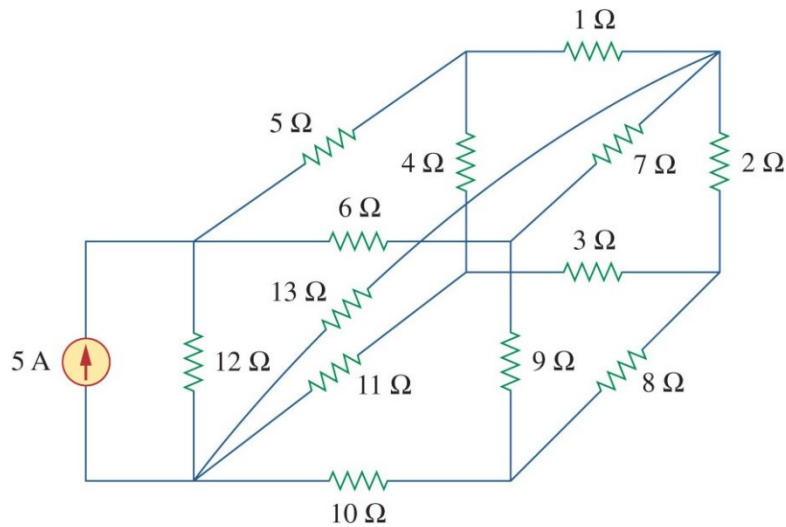


This is a planar circuit: It can be redrawn to avoid crossing branches



This is a nonplanar circuit: The branch with the 13 Ω resistor prevents the circuit from being drawn without crossing branches

Non-Planar Circuit

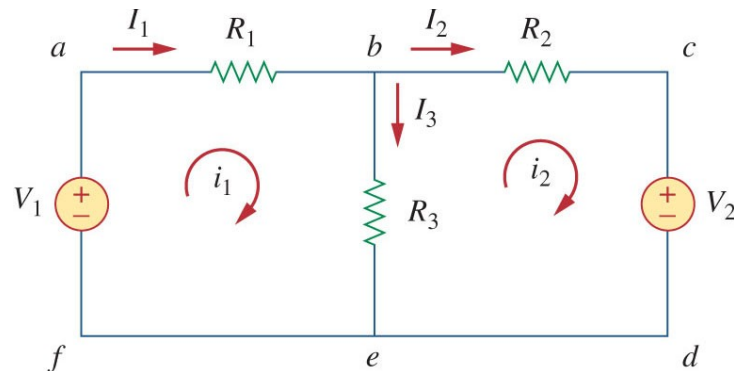


MA Steps

Mesh analysis follows these steps:

1. Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes
2. Apply KVL to each of the n mesh currents
3. Solve the resulting n simultaneous equations to get the mesh currents

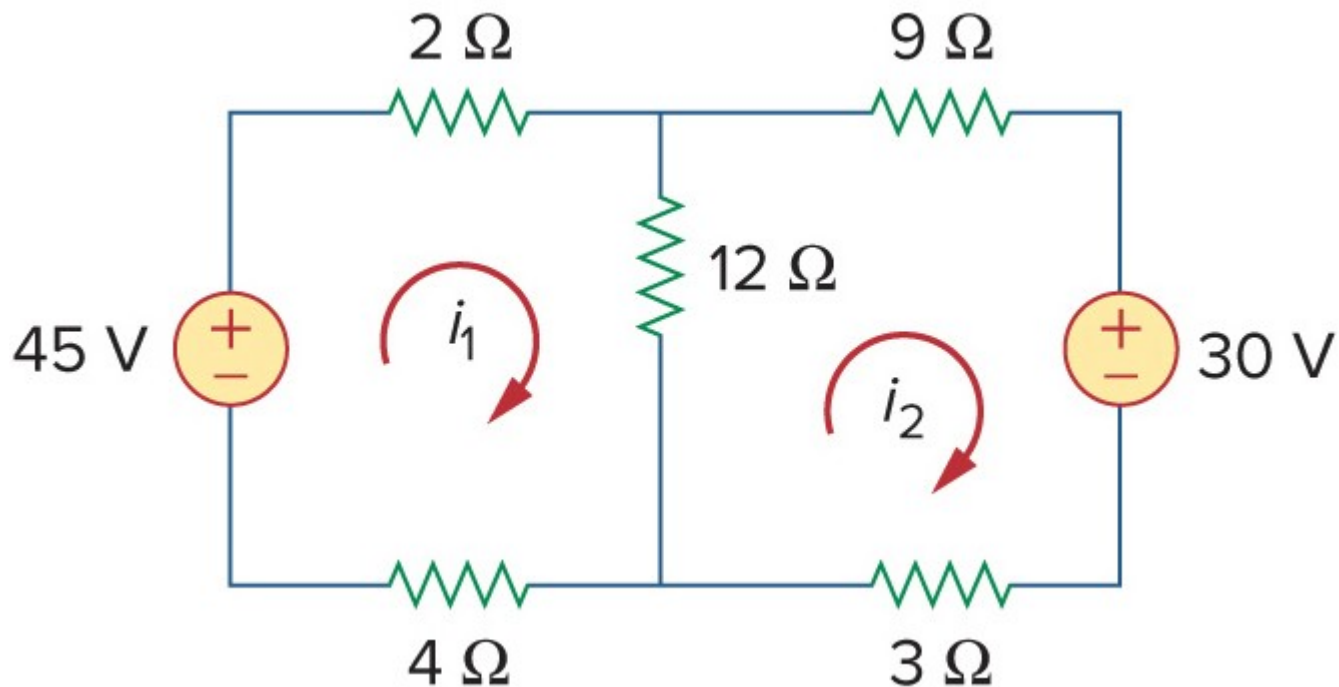
MA Example



- The above circuit has two paths that are meshes (abefa and bcdeb)
- The outer loop (abcdefa) is a loop, but not a mesh
- First, mesh currents i_1 and i_2 are assigned to the two meshes then KVL is applied to them

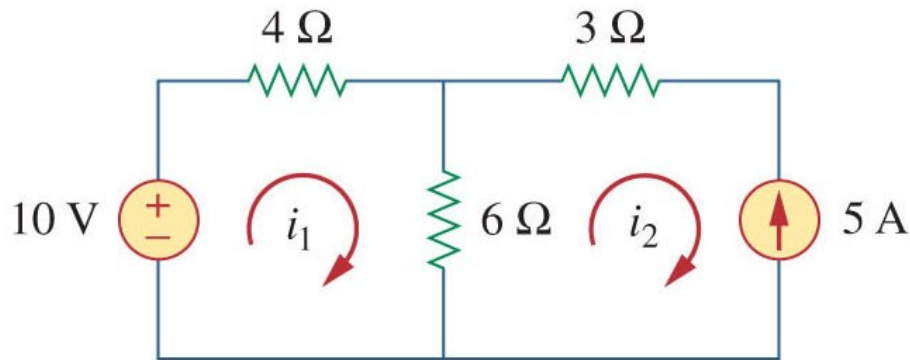
Practice Problems

3.5 Find i_1 and i_2



MA with Current Sources

- The presence of a current source makes the mesh analysis simpler in that it reduces the number of equations
- If the current source is located on only one mesh, the current for that mesh is defined by the source
- For example:

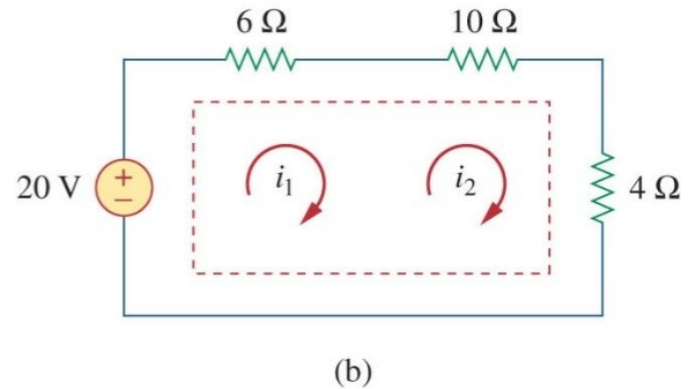
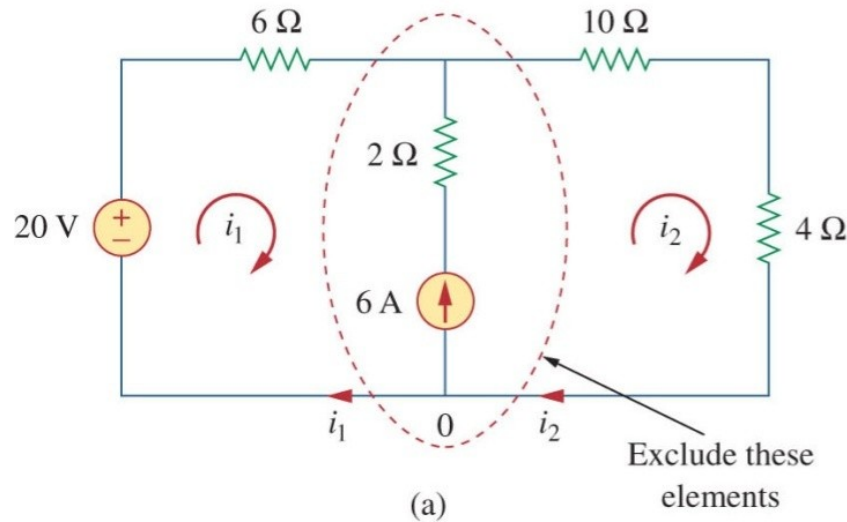


- Here, the current i_2 is equal to -5A

Supermesh

- Similar to the case of NA (with V-sources), I-sources shared by more than one mesh need special treatment – they must be joined together into a "supermesh"
- The supermesh is constructed by merging the two meshes and excluding the shared source and any elements in series with it
- A supermesh is required because mesh analysis uses KVL, but we don't know the V across a I-source in advance
- Note: intersecting supermeshes in a circuit must be combined to form a larger supermesh

Creating a Supermesh



- In this example, a 6A current source is shared between mesh 1 and 2
- The supermesh is formed by merging the two meshes
- The current source and the 2Ω resistor in series with it are removed

Equations to Solve

- Apply KVL to the supermesh

$$-20 + 6i_1 + 10i_2 + 4i_2 = 0 \Rightarrow$$

$$6i_1 + 14i_2 = 20$$

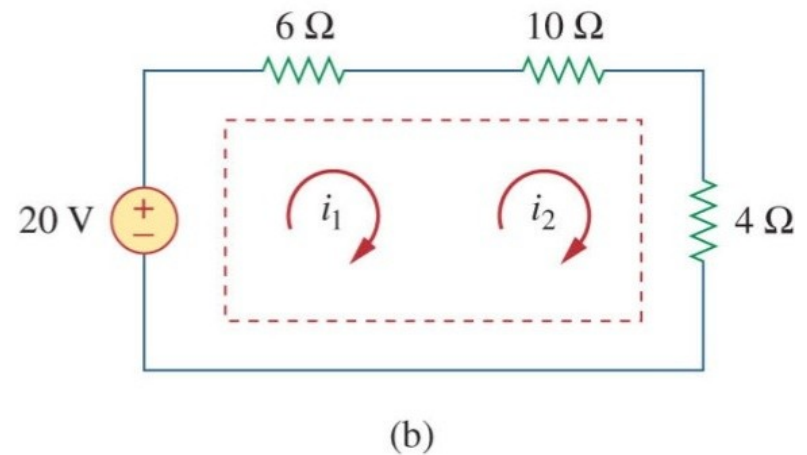
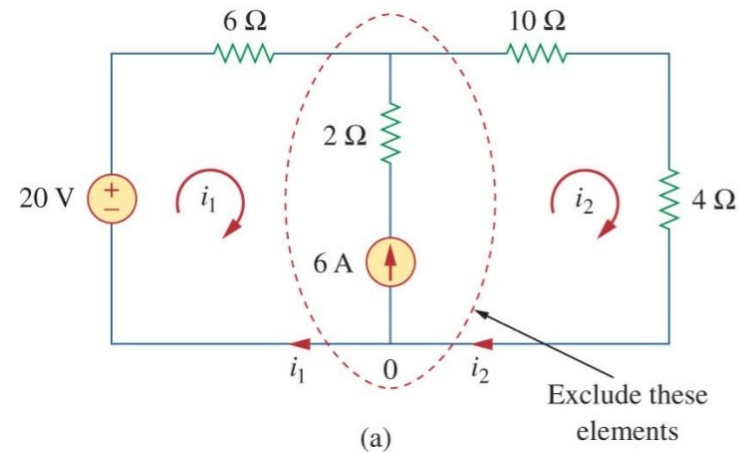
- Next apply KCL to the node in the branch where the two meshes intersect

$$i_1 + 6 - i_2 = 0 \Rightarrow$$

$$i_1 - i_2 = -6$$

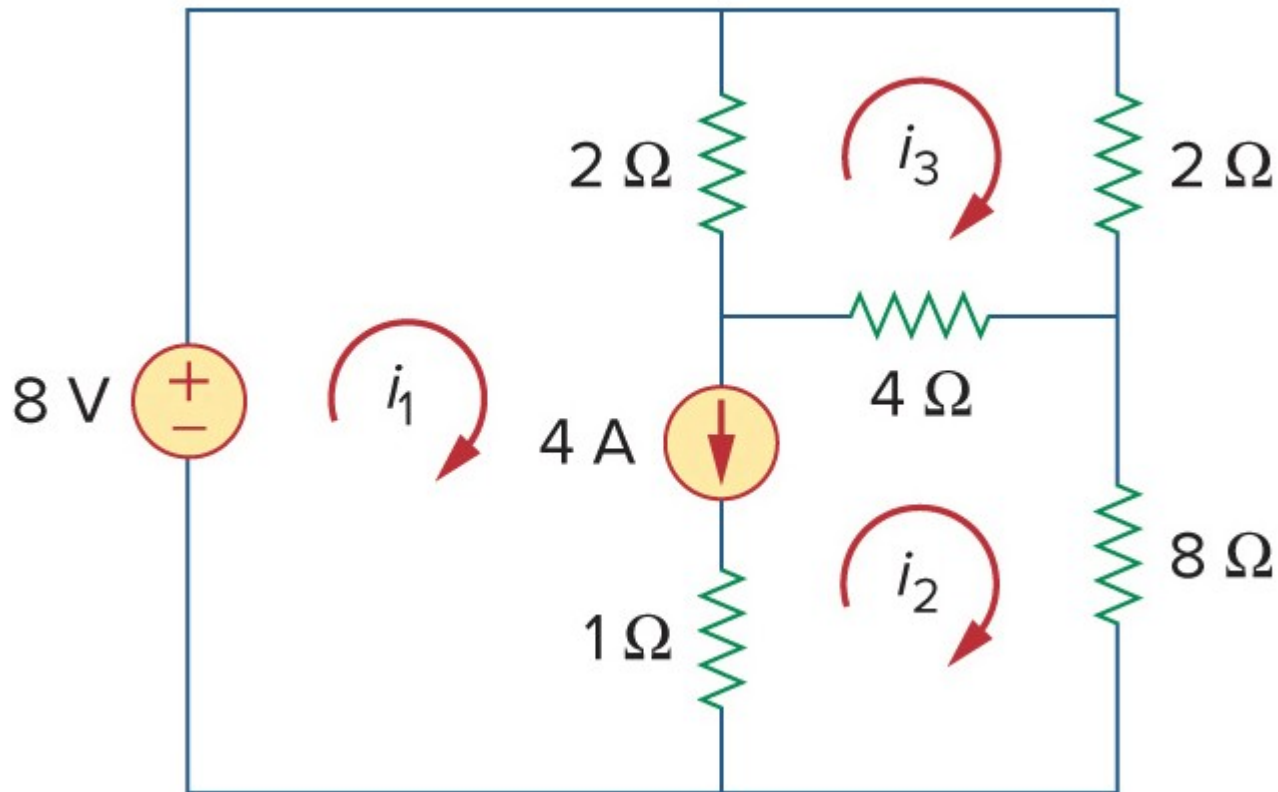
- Solve simultaneous equations

$$i_1 = -3.2 \text{ A} \quad i_2 = 2.8 \text{ A}$$



Practice Problems

3.7 Find i_1 , i_2 and i_3



Nodal Analysis by Inspection

- There is a faster way to construct a matrix for solving a circuit by nodal analysis
- It requires that all current sources within the circuit be independent
- In general, for a circuit with N non-reference nodes, the node-voltage equations may be written as:

$$\begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1N} \\ G_{21} & G_{22} & \cdots & G_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ G_{N1} & G_{N2} & \cdots & G_{NN} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix}$$

Nodal Analysis by Inspection

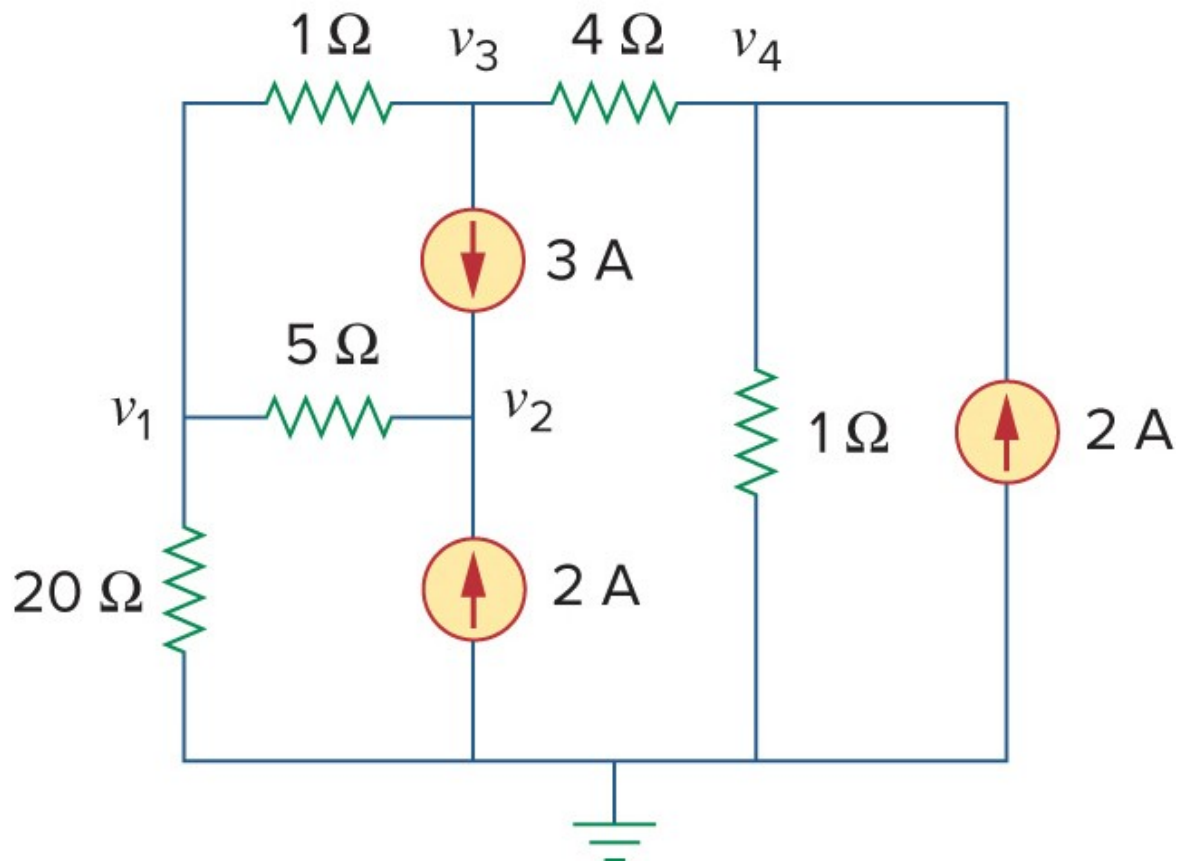
- Each diagonal term on the conductance matrix is the sum of conductances connected to the node indicated by the matrix index
- The off diagonal terms, G_{jk} are the negative of the sum of all conductances connected between nodes j and k with $j \neq k$

Nodal Analysis by Inspection

- The unknown voltages are denoted as v_k
- The sum of all independent current sources directly connected to node k are denoted as i_k
Current entering the node are treated as positive and vice versa
- This matrix equation can be solved for the unknown values of the nodal voltages

Practice Problems

3.8 Find node voltages by inspection



Mesh Analysis by Inspection

- There is a similarly fast way to construct a matrix for solving a circuit by mesh analysis
- It requires that all voltage sources within the circuit be independent
- In general, for a circuit with N meshes, the mesh-current equations may be written as:

$$\begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1N} \\ R_{21} & R_{22} & \cdots & R_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ R_{N1} & R_{N2} & \cdots & R_{NN} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

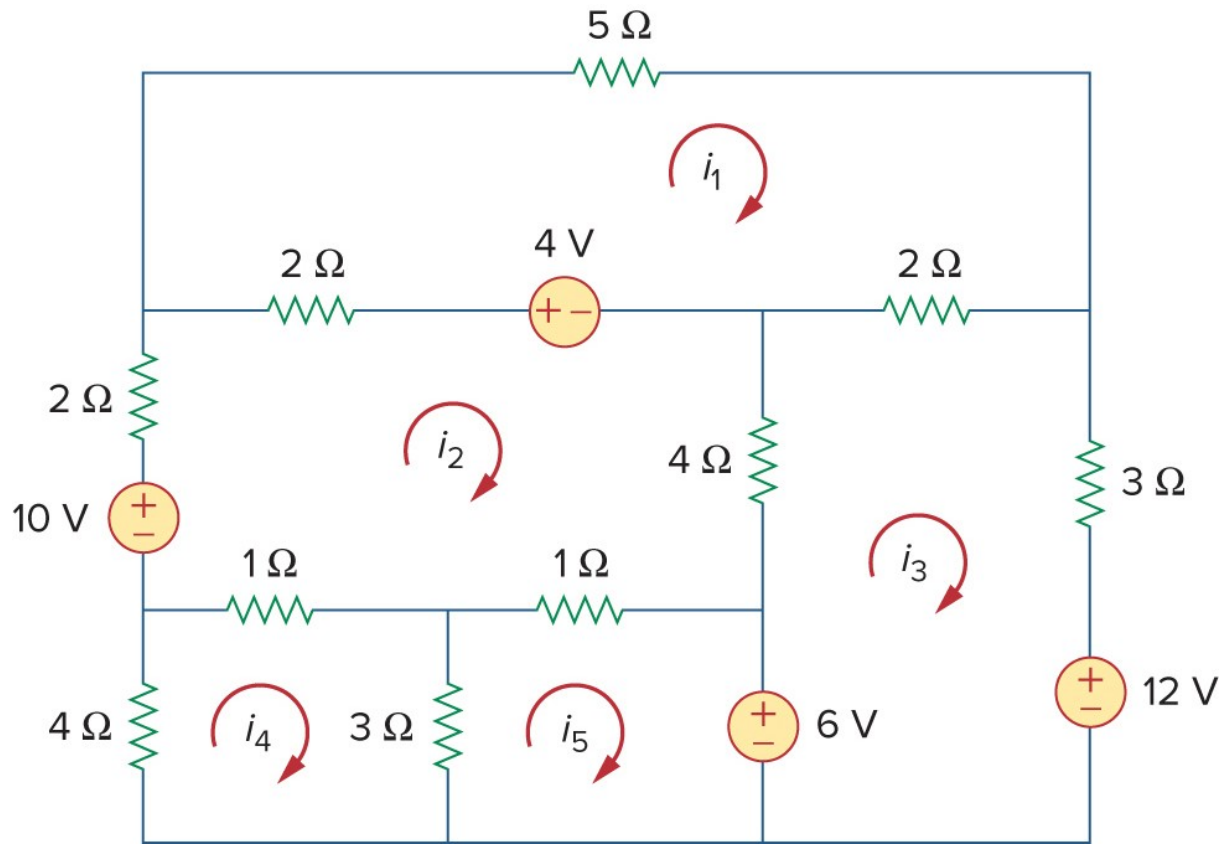
- Each diagonal term on the resistance matrix is the sum of resistances in the mesh indicated by the matrix index

Mesh Analysis by Inspection

- The off diagonal terms, R_{jk} are the negative of the sum of all resistances in common with meshes j and k with $j \neq k$
- The unknown mesh currents in the clockwise direction are denoted as i_k
- The sum taken clockwise of all voltage sources in mesh k are denoted as v_k with voltage rises treated as positive
- This matrix equation can be solved for the values of the unknown mesh currents

Practice Problems

3.9 Find mesh currents by inspection



Selecting an Appropriate Approach

In principle both the nodal analysis and mesh analysis are useful for any given circuit

What then determines if one is going to be more efficient for solving a circuit problem?

There are two factors that dictate the best choice:

- The nature of the particular network is the first factor
- The second factor is the information required

Mesh analysis when...

- ...the network contains:
 - Many series connected elements
 - Voltage sources
 - Supermeshes
 - A circuit with fewer meshes than nodes
- Note: mesh analysis is the only suitable analysis for transistor circuits
- It is not appropriate for operational amplifiers because there is no direct way to obtain the voltage across an op-amp

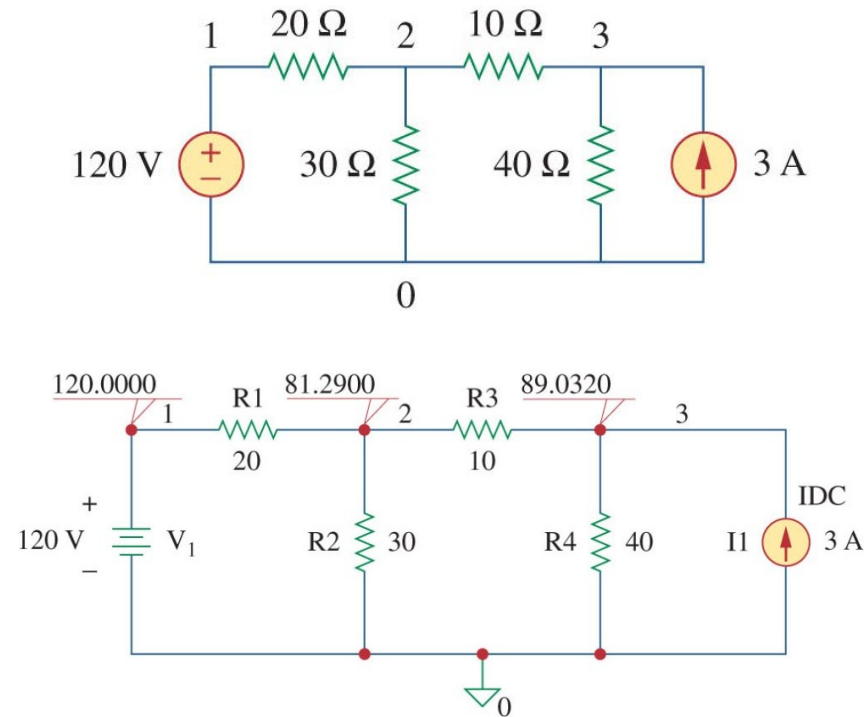
Nodal analysis if...

- ...the network contains:
 - Many parallel connected elements
 - Current sources
 - Supernodes
 - Circuits with fewer nodes than meshes
- Note: Non-Planar circuits can only be solved using nodal analysis
- This format is easier to solve by computer

Circuit Analysis with Spice

- LTSpice (et al) is a common program used for circuit analysis.
- It is capable of determining all of the branch voltages and currents if the numerical values for all circuit components are known.
- Analysis using LTSpice begins with drawing a schematic view of the circuit.
- The node voltages of interest can be indicated during the schematic setup using 'VIEWPOINTS.'
- The values are obtained by running 'Analysis/Simulate.'

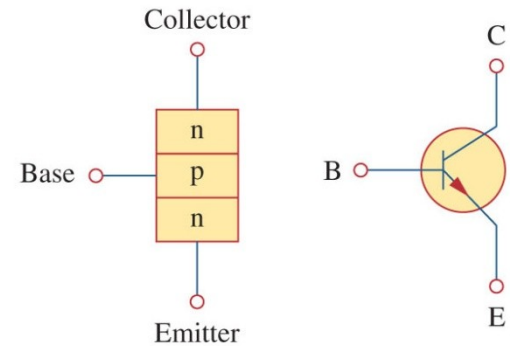
Rendering a circuit in LTSpice



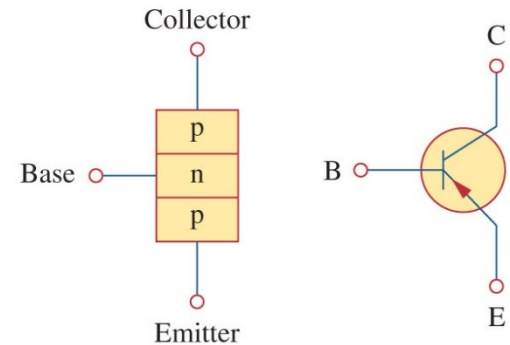
Converting a standard schematic (top) to a Spice schematic (bottom) is fairly straightforward. Note the explicit definition of the reference node by using the ground symbol labeled with a '0' and the nodal voltages of interest displayed with the 'viewpoints'

Application: DC transistor circuit

- Here we will use the approaches learned in this chapter to analyze a transistor circuit
- In general, there are two types of transistors commonly used: Field Effect (FET) and Bipolar Junction (BJT)
- This problem will use a BJT
- A BJT is a three terminal device, where the input current into one terminal (the base) affects the current flowing out of a second terminal (the collector)
- The third terminal (the emitter) is the common terminal for both currents



(a)



(b)

KCL and KVL for a BJT

- The currents from each terminal can be related to each other as follows:

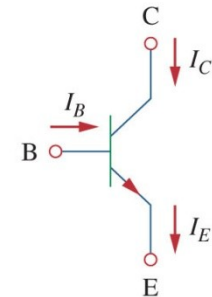
$$I_E = I_B + I_C$$

- The base and collector current can be related to each other by the parameter β , which can range from 50-1000:

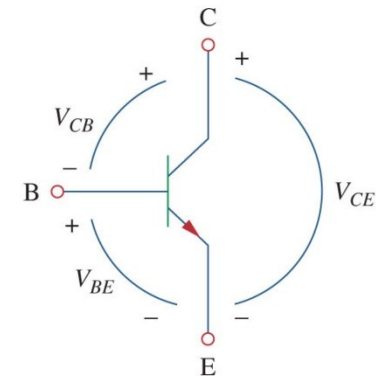
$$I_C = \beta I_B$$

- Applying KVL to the BJT gives:

$$V_{CE} + V_{EB} + V_{BC} = 0$$



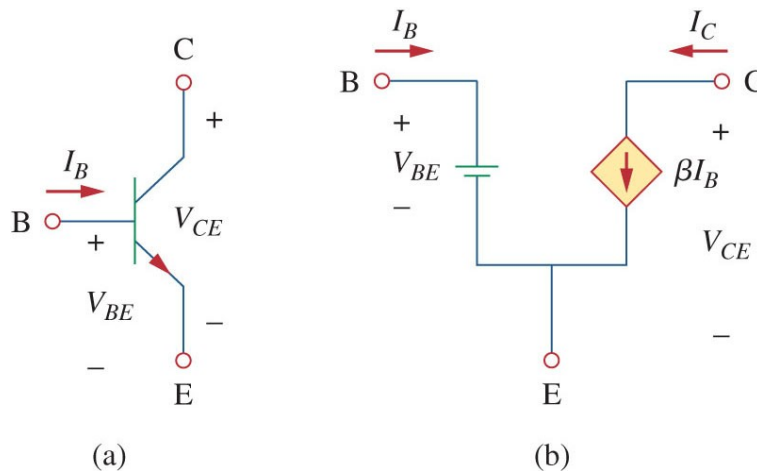
(a)



(b)

DC model of a BJT

- A transistor has a few operating modes depending on the applied voltages/currents. In this problem, we will be interested in the operation in “active mode”
- This is the mode used for amplifying signals
- The figure below shows the equivalent DC model for a BJT in active mode

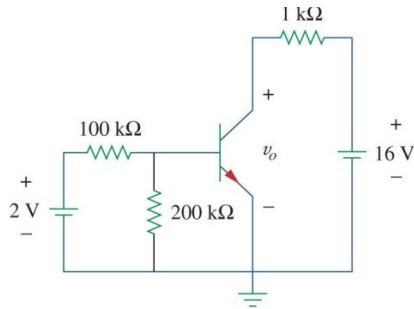


- Note that nodal analysis can only be applied to the BJT after using this model

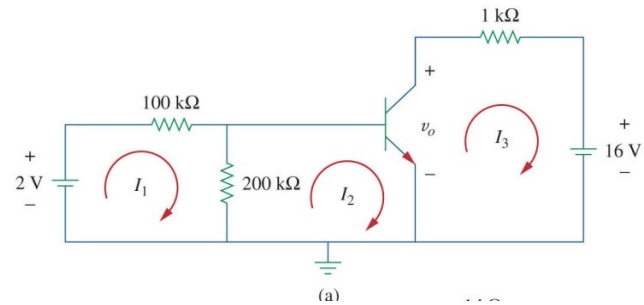
Setting up a BJT circuit

Below are three approaches to solving a transistor circuit

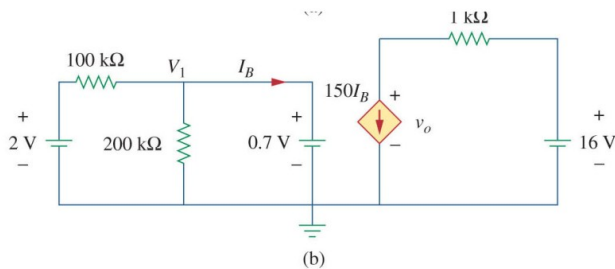
Note when the equivalent model is used and when it is not



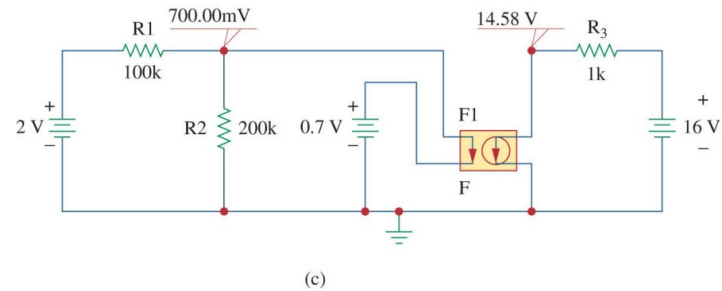
Original circuit



Mesh analysis



Nodal analysis



Spice analysis

Practice Problems

3.12 Let $\beta=100$ and $V_{BE}=0.7\text{V}$. Determine v_o and V_{CE}

